

Important information. Please read carefully.

## ***GLENIX***

Tablet 2mg

### **COMPOSITION**

GLENIX Tablet 2mg: Each tablet contains Glimepiride 2mg

### **PHARMACODYNAMICS**

Glimepiride is a blood glucose lowering agent belonging to the sulphonylurea group. The decrease in blood glucose is achieved principally by stimulating insulin release from pancreatic beta cells. This effect is predominantly based on improved responsiveness of these cells to the physiological glucose stimulus. Glimepiride augments the normal action of insulin on peripheral glucose uptake. Moreover, it mimics such action as well as the glucose output of the liver.

Good metabolic control over 24 hours can be achieved with a single daily dose. In patients with insufficient response to the maximum daily dose, combined use with an additional oral antidiabetic containing metformin or with insulin improves metabolic control.

### **PHARMACOKINETICS**

**Absorption:** The bioavailability of glimepiride after oral administration is complete. Food intake has no relevant influence on absorption, only absorption rate is slightly diminished. Maximum serum concentrations ( $C_{max}$ ) are reached approx. 2.5 hours after oral intake (mean 0.3µg/ml during multiple dosing of 4mg daily) and there is a linear relationship between dose and both  $C_{max}$  and AUC (area under the time/concentration curve).

**Distribution:** Glimepiride has a very low distribution volume (approx. 8.8 litres) which is roughly equal to the albumin distribution space, high protein binding (>99%), and a low clearance (approx. 48 ml/min).

**Biotransformation and elimination:** Mean dominant serum half-life, which is of relevance for the serum concentrations under multiple-dose conditions, is about 5 to 8 hours. After high doses, slightly longer half-lives were noted.

After a single dose of radiolabelled glimepiride, 58% of the radioactivity was recovered in the urine, and 35% in the faeces. No unchanged substance was detected in the urine. Two metabolites, the hydroxy derivative and the carboxy derivative, which most probably resulting from hepatic metabolism (major enzyme is CYP2C9) were identified both in urine and faeces. After oral administration of glimepiride, the terminal half-lives of these metabolites were 3 to 6 and 5 to 6 hours respectively.

Comparison of single and multiple once-daily dosing revealed no significant differences in pharmacokinetics, and the intra individual variability was very low. There was no relevant accumulation.

Pharmacokinetics were similar in males and females, as well as in young and elderly (above 65 years) patients. In patients with low creatinine clearance, there was a tendency for glimepiride clearance to increase and for average serum concentrations to decrease, most probably resulting from a more rapid elimination because of lower protein binding. Renal elimination of the two metabolites was impaired. Overall no additional risk of accumulation is to be assumed in such patients.

## **INDICATIONS**

As an adjunct to diet and exercise in non-insulin-dependent (type II) diabetes, when diet, physical exercise and weight reduction alone are not adequate. Glenix may also be used in combination with an oral antidiabetic containing metformin or with insulin.

## **DOSAGE AND ADMINISTRATION**

In principle, the dosage of Glenix is governed by the desired blood sugar levels. The dosage of Glenix must be the lowest which is sufficient to achieve the desired metabolic control.

Treatment with Glenix must be initiated and monitored by a doctor. Glenix must be taken at the times and in the doses prescribed. If a dose is forgotten, this should not be corrected by increasing the next dose. Measures for dealing with mistakes (in particular forgetting a dose or skipping a meal) or situations where a dose cannot be taken at the prescribed time must be discussed and agreed between doctor and patient beforehand. A doctor must be notified immediately if the dose taken is too high, or an extra dose has been taken.

The initial and maintenance doses are set based on the results of regular checks of blood and urine glucose. Monitoring of glucose levels in blood and urine also serves to detect either primary or secondary failure of therapy.

Initial dose and dose titration: The usual initial dose is 1mg Glenix once daily. If control is unsatisfactory the dosage should be increased, based on the glycaemic control, in a stepwise manner with an interval of about 1 to 2 weeks between each step, to 2, 3 or 4mg Glenix per day.

A dosage of more than 4mg Glenix per day gives better results only in exceptional cases. The maximum recommended dose is 6mg Glenix per day.

Distribution of doses: Normally a single daily dose of Glenix is sufficient. It is recommended that this dose be taken shortly before or during a substantial breakfast or if none is taken - shortly before or during the first main meal. It is very important not to skip meals after taking Glenix.

Secondary dosage adjustment: In the course of treatment, as an improvement in control of diabetes is associated with higher insulin sensitivity, Glenix requirements may fall. To avoid hypoglycaemia, timely dose reduction or cessation of therapy must therefore be considered. Change in dosage may also be necessary, if there are changes in weight or life style of the patient, or other factors that increase the risk of hypo- or hyperglycaemia.

Duration of treatment: Treatment with Glenix is normally a long-term treatment.

Changeover from other oral antidiabetics to Glimepiride: There is no exact dosage relationship between Glenix and other oral antidiabetics. When substituting Glenix for other such agents, the initial daily dose is 1mg; this applies even in changeovers from the maximum dose of another oral blood glucose lowering agent. Any Glenix dose increase should be in increase “Initial dose and dose titration”.

Consideration must be given to the potency and duration of action of the previous antidiabetics. It may be necessary to interrupt treatment to avoid additive effects which would increase the risk of hypoglycaemia.

Use in combination with insulin: In patients not adequately controlled with the maximum daily dose of Glenix, concomitant insulin therapy can be initiated if necessary. While maintaining the Glenix dose, insulin treatment is started at low dose and titrated up depending on the desired level of metabolic control. The therapy should be initiated under close medical supervision.

Tablets should be swallowed without chewing and with sufficient amount of liquid (approximately  $\frac{1}{2}$  glass).

## **CONTRAINDICATIONS**

Glimepiride should not be used in the following cases: insulin dependent (type 1) diabetes mellitus, diabetic precoma or coma, ketoacidosis, severe renal or hepatic function disorders, hypersensitivity to glimepiride, other sulphonylureas or sulphonamides or excipients in the tablet.

In case of severe renal or hepatic function disorders, a change over to insulin is required.

## **WARNINGS AND PRECAUTIONS**

Glimepiride must be taken shortly before or during a meal. When meals are taken at irregular hours or skipped altogether, treatment with glimepiride may lead to hypoglycaemia.

Factors favouring hypoglycaemia include:

- unwillingness or (more commonly in older patients) incapacity of the patient to cooperate,

- under nutrition, irregular mealtimes or missed meals or periods of fasting,
- alterations of diet,
- imbalance between physical exertion and carbohydrate intake,
- consumption of alcohol, especially in combination with skipped meals,
- impaired renal function,
- severe liver dysfunction,
- overdosage with glimepiride,
- certain uncompensated disorders of the endocrine system affecting carbohydrate metabolism or counter regulation of hypoglycaemia (for example in certain disorders of thyroid function and in anterior pituitary or adrenocortical insufficiency),
- concurrent administration of certain other medicines (see “Drug Interactions”).

The doctor must be informed about such factors and about hypoglycaemic episodes, since these require particularly careful monitoring.

If such risk factors for hypoglycaemia are present, it may be necessary to adjust the dosage of glimepiride or the entire therapy. This also applies whenever illness occurs during therapy or the patient’s life-style changes.

Those symptoms of hypoglycaemia which reflect the body’s adrenergic counter-regulation (see “Side Effects”) may be milder or absent in those situations where hypoglycaemia develops gradually, in the elderly, and in patients with a certain type of nervous disease (autonomic neuropathy) or those receiving concurrent treatment with beta-blockers, clonidine, reserpine, guanethidine, or other sympatholytic medicines.

Hypoglycaemia can almost always be promptly controlled by immediate intake of sugar, e.g. in the form of glucose, sugar cubes or sugar-sweetened beverages. Patients should always carry at least 20 grams of glucose with them for this purpose (food or beverages containing artificial sweeteners- such as diet foods or drinks- are ineffective in controlling hypoglycaemia).

It is known from other sulfonylureas that, despite initially successful countermeasures, hypoglycaemia may recur. Therefore, continued close observation is necessary. Severe hypoglycaemia requires, in addition, immediate treatment and follow-up by a doctor and, in some circumstances, hospitalisation.

If treated by different doctors (upon, e.g., admission to hospital after an accident, illness while on holiday), the patients must inform them about their diabetes and previous treatment.

In exceptional stress situations (e.g. trauma, surgery, infections with fever) blood sugar control may deteriorate, and a temporary change to insulin may be necessary.

During treatment with glimepiride, glucose levels in blood and urine must be checked regularly, as should, additionally, the proportion of glycated haemoglobin.

Alertness and reactions may be impaired due to hypo- or hyperglycaemia, especially when beginning or after altering treatment, or when glimepiride is not taken regularly. Such impairment may, for example, affect the ability to operate a vehicle or machinery.

No experience has been gained concerning the use of glimepiride in patients with severe impairment of liver function or dialysis patients. In patients with severe impairment of renal or liver function change over to insulin is indicated.

#### Special Warning On Increased Risk Of Cardiovascular Mortality.

The administration of oral hypoglycemic drugs has been reported to be associated with increased cardiovascular mortality as compared to treatment with diet alone or diet plus insulin. This warning is based on the study conducted by the University Group Diabetes Program (UGDP), a long-term, prospective clinical trial designed to evaluate the effectiveness of glucose-lowering drugs in preventing or delaying vascular complications in patients with non-insulin-dependent diabetes. The study involved 823 patients who were randomly assigned to one of four treatment groups (Diabetes, 19 supp. 2: 747-830, 1970).UGDP reported that patients treated for 5 to 8 years with diet plus a fixed dose of tolbutamide (1.5 grams per day) had a rate of cardiovascular mortality approximately 2-1/2 times that of patients treated with diet alone. A significant increase in total mortality was not observed, but the use of tolbutamide was discontinued based on the increase in cardiovascular mortality, thus limiting the opportunity for the study to show an increase in overall mortality. Despite controversy regarding the interpretation of these results, the findings of the UGDP study provide an adequate basis for this warning. The patient should be informed of the potential risks and advantages of GLIMEPIRIDE (glimepiride tablets) and of alternative modes of therapy.

Although only one drug in the sulfonylurea class (tolbutamide) was included in this study, it is prudent from a safety standpoint to consider that this warning may also apply to other oral hypoglycemic drugs in this class, in view of their close similarities in mode of action and chemical structure.

#### **DRUG INTERACTIONS**

Patients who take or discontinue taking certain other medicines while undergoing treatment with glimepiride may experience changes in blood sugar control.

Based on the experience with glimepiride and on what is known of other sulphonylureas, the following interactions must be considered:

Potential of the blood glucose lowering effect and, thus, in some instances hypoglycaemia may occur when one of the following drugs is taken, for example: insulin and other oral antidiabetics, ACE inhibitors, allopurinol, anabolic steroids and male sex hormones, chloramphenicol, coumarin derivatives, cyclophosphamide, disopyramide, fenfluramine, fenyramidol, fibrates, fluoxetine, guanethidine, ifosfamide, MAO inhibitors, miconazole, para-aminosalicylic acid, pentoxifylline (high dose parenteral), phenylbutazone, azapropazone, oxyphenbutazone, probenecid, quinolones, salicylates, sulfinpyrazone, sulfonamides, tetracyclines, tritoqualine, trofosfamide.

Weakening of the blood glucose lowering effect and, thus raised blood glucose levels may occur when one of the following drugs is taken, for example: acetazolamide, barbiturates, corticosteroids, diazoxide, diuretics, epinephrine (adrenaline) and other sympathomimetic agents, glucagon, laxatives (after protracted use), nicotinic acid (in high doses), oestrogens and progestogens, phenothiazines, phenytoin, rifampicin, thyroid hormones.

H<sub>2</sub> receptor antagonists, clonidine and reserpine may lead to either potentiation or weakening of the blood glucose lowering effect.

Beta-blockers decrease glucose tolerance. In patients with diabetes mellitus, this may lead to deterioration of metabolic control. In addition, beta-blockers may increase the tendency to hypoglycaemia (due to impaired counter-regulation)

Under the influence of sympatholytic drugs such as betablockers, clonidine, guanethidine and reserpine, the signs of adrenergic counter regulation to hypoglycaemia may be reduced or absent.

Both acute and chronic alcohol intake may potentiate or weaken the hypoglycaemic action of glimepiride unpredictably.

The effect of coumarin derivatives may be potentiated or weakened.

## **PREGNANCY AND LACTATION**

Glimepiride is contraindicated in pregnancy, a changeover to insulin is necessary. Patients planning a pregnancy must inform their doctor, and should changeover to insulin. Glimepiride must not be taken by breast feeding women, and a changeover to insulin or discontinuation of breast feeding is necessary.

## **SIDE EFFECTS**

Hypoglycaemia: As a result of the blood-glucose lowering action of glimepiride, hypoglycaemia may occur, and may be prolonged.

Possible symptoms of hypoglycaemia include: headache, ravenous hunger, nausea, vomiting, lassitude, sleepiness, disordered sleep, restlessness, aggressiveness, impaired concentration, alertness and reactions, depression, confusion, difficulty in speaking and even speech loss, visual disorders, tremor, pareses, sensory disturbances, dizziness, helplessness, loss of self-control, delirium, cerebral convulsions, somnolence and loss of consciousness up to and including coma, shallow respiration and bradycardia. In addition, signs of adrenergic counter-regulation may be present such as sweating, clammy skin, anxiety, tachycardia, hypertension, palpitations, angina pectoris and cardiac arrhythmias. The clinical picture of a severe hypoglycaemic attack may resemble that of a stroke. The symptoms of hypoglycaemia nearly always subside when hypoglycaemia is corrected.

Eyes: Transient visual disturbances may occur especially on initiation of treatment, due to changes in blood glucose levels.

Gastrointestinal: Gastrointestinal complaints like nausea, vomiting and diarrhoea, pressure or a feeling of fullness in the stomach and abdominal pain are very rare and seldom lead to discontinuation of therapy.

Hepato-biliary: Elevation of liver enzymes may occur. In very rare cases, impairment of liver function (e.g. with cholestasis and jaundice) may develop, as well as hepatitis which may progress to liver failure.

Blood and lymphatic system: Changes in haematology are rare during glimepiride treatment. Moderate to severe thrombocytopenia, leucopenia, erythrocytopenia, granulocytopenia, agranulocytosis, haemolytic anaemia and pancytopenia may occur. These are in general reversible upon discontinuation of medication.

Skin and subcutaneous tissue: Hypersensitivity reactions of the skin may occur as itching, rash and urticaria. In very rare cases hypersensitivity to light may occur.

Immune system: In very rare cases mild hypersensitivity reactions may develop into serious reactions with dyspnoea, fall in blood pressure and sometimes shock. Allergic vasculitis is possible in very rare cases. Cross allergenicity with sulphonylureas, sulphonamides or related substances is possible.

Investigations: In very rare cases, a decrease in the sodium serum concentrations may occur.

## **SYMPTOMS AND TREATMENT FOR OVERDOSAGE**

After ingestion of an overdose, hypoglycaemia may occur, lasting from 12 to 72 hours, and may recur after an initial recovery. Symptoms may not be present for up to 24 hours after ingestion. In general observation in hospital is recommended. Nausea, vomiting and epigastric pain may occur. The hypoglycaemia may in general be accompanied by neurological symptoms like restlessness, tremor, visual disturbances, co-ordination problems, sleepiness, coma and convulsions.

Treatment primarily consists of preventing absorption by inducing vomiting and then drinking water or lemonade with activated charcoal (adsorbent) and sodium-sulphate (laxative). If large quantities have been ingested, gastric lavage is indicated, followed by activated charcoal and sodium-sulphate. In case of (severe) overdose hospitalisation in an intensive care department is indicated. Start the administration of glucose as soon as possible, if necessary by a bolus intravenous injection of 50 ml of a 50% solution, followed by an infusion of a 10% solution with strict monitoring of blood glucose. Further treatment should be symptomatic.

In particular when treating hypoglycaemia due to accidental intake of glimepiride in infants and young children, the dose of glucose given must be carefully controlled to avoid the possibility of producing dangerous hyperglycaemia. Blood glucose should be closely monitored.

**STORAGE CONDITION**

Store below 25°C.

**SHELF LIFE**

3 years from the date of manufacture.

**PRODUCT DESCRIPTION**

Light green oval shape normal convex tablet, engraved with “XS” logo and break line on one side and plain on the reverse side. Available in PVdC/Alu blister strips of 10’s in packing of 30 tablets per box.

**KEEP OUT OF REACH OF CHILDREN  
JAUHI DARI KANAK-KANAK****MANUFACTURED AND MARKETING BY**

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For further information, please consult your pharmacist or physician.

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